

This assignment was a bit easier than the previous one. In this assignment, I loaded the Titanic dataset and performed an initial inspection using `".head()"` and `".info()"`.

Although I ran into some errors, mostly related to syntax. For example: I fixed a `SyntaxError` in the import statements to ensure all necessary libraries were loaded correctly.

I generated summary statistics for the numerical columns using `".describe()"` and extracted key insights, such as the average age of passengers and the overall survival rate.

In order to get a better picture of the events that transpired. I decided to explore several relationships through visualizations, and used a `countplot` to see the overall distribution of survivors vs. non-survivors.

I also used a `countplot` to show a strong correlation between passenger class and survival rate, with 1st class passengers having a much higher chance of survival.

I Used a `countplot` to demonstrate the significant difference in survival rates between females and males, supporting the 'women and children first' narrative.

I Used a `FacetGrid` with histograms to visualize age distributions for survivors and non-survivors, highlighting better survival rates for young children.

Generated a `countplot` to analyze survival rates based on the embarkation port, noting that passengers from Cherbourg (C) had proportionally higher survival.

Fare vs. Survival: Used a `boxplot` to illustrate that passengers who paid higher fares generally had a better chance of survival, indicating a link between fare (and implicitly, class) and survival.

I made sure to answer all reflection questions, explaining the primary goal of EDA, identifying the characteristics of a person with the best chance of survival, and discussing the importance of data visualization over summary statistics alone.

Overall this was a fun and quick assignment, i learnt a lot about visualization and a small dive into the history of the Titanic

## **Reflection Journal**

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### **Exploratory Data Analysis**

During this assignment, I gained a better understanding of both the technical and collaborative aspects of completing a machine learning project as a team. Working through the lab helped me become more comfortable using Jupyter Notebook, reviewing code and outputs, organizing project files, and using GitHub as a collaborative platform.

One of the biggest things I learned from this assignment was the importance of organization and communication when multiple people are contributing to the same project. Each team member completed individual portions of the work, so keeping track of files and contributions was necessary to make sure the final project could be compiled correctly.

I also gained more experience working with GitHub repositories and the pull request workflow. This helped me understand how version control can be used to collect contributions while maintaining an organized record of the team's work.

Overall, this assignment strengthened my technical skills while also showing me how important clear procedures, communication, and individual accountability are when working on a collaborative technical project.